

Seongjun Yang

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Research Interests

I develop AI methods that move beyond black-box prediction to uncover interpretable, mechanistic insights from genomic data. My interests include regulatory genomics, genomic foundation models, model robustness, and evidence-grounded evaluation, with an emphasis on generating testable biological hypotheses for precision medicine and longevity healthcare.

Education

Cold Spring Harbor Laboratory

Ph.D. Student

Cold Spring Harbor, NY

2026 – Present

Advisor: Prof. Peter Koo. Research in interpretable and robust AI for genomics.

Korea Advanced Institute of Science and Technology (KAIST)

M.S. in Artificial Intelligence

South Korea

Sep 2020 – Aug 2022

Advisor: Prof. Edward Choi. Thesis: *Towards the Practical Utility of Federated Learning in the Medical Domain*.

Yonsei University

B.S. in Computer Science, Magna Cum Laude

South Korea

Mar 2014 – Aug 2020

Department Prizes for Outstanding Student Performance; full scholarship (2017–2019).

Ongoing Research

Under review **PBISC: Reference-free Single-cell-resolution Deconvolution of PBMC Bulk RNA-seq**.

Under review **RNA-LLM: Transcriptomes as One Language-model Token**.

Selected Publications

- 2026 Kyunghoon Hur, Eunjung Jeon, Hyun Woo Kim, Gyubok Lee, and **Seongjun Yang**. *Leveraging Biokinetic Knowledge Priors for Data-Scarce Bioprocess Modeling*. ICML AI for Science Workshop.
- 2025 Suhyeong Kim, **Seongjun Yang**, and Seongwoo Kim. *Assessing Patent Value with LLMs: DRAM Dataset*. PACIS.
- 2024 **Seongjun Yang***, Gibbeum Lee*, Jaewoong Cho, Dimitris Papailiopoulos, and Kangwook Lee. *Predictive Pipelined Decoding: A Compute-Latency Trade-off for Exact LLM Decoding*. TMLR.
- 2023 **Seongjun Yang***, Hyeonji Hwang*, Daeyoung Kim, Radhika Dua, Jong-Yeup Kim, Eunho Yang, and Edward Choi. *Towards the Practical Utility of Federated Learning in the Medical Domain*. CHIL.
- 2023 Radhika Dua, **Seongjun Yang**, Yixuan Li, and Edward Choi. *Task Agnostic and Post-hoc Unseen Distribution Detection*. WACV.
- 2022 Gyubok Lee, Hyeonji Hwang, Seongsu Bae, Yeonsu Kwon, Woncheol Shin, **Seongjun Yang**, Minjoon Seo, Jong-Yeup Kim, and Edward Choi. *EHRSQL: A Practical Text-to-SQL Benchmark for Electronic Health Records*. NeurIPS Datasets and Benchmarks.
- 2022 Junu Kim, Kyunghoon Hur, **Seongjun Yang**, and Edward Choi. *Universal EHR Federated Learning Framework*. ML4H Extended Abstract.
- 2021 Gyubok Lee*, **Seongjun Yang***, and Edward Choi. *Improving Lexically Constrained Neural Machine Translation with Source-Conditioned Masked Span Prediction*. ACL-IJCNLP (Short Papers).

* Equal contribution.

Professional Experience

Gruve AI United States (Remote)
AI Researcher Dec 2025 – May 2026

- Developed AI agents in collaboration with a genetic testing company.

KRAFTON AI South Korea
NLP Research Engineer Nov 2022 – 2025

- Researched LLM post-training, efficient inference, model compression, and language-model applications for games.
- Developed instruction-tuning and prompting strategies for large language models.
- Projects included Predictive Pipelined Decoding, KORani, and AutoEvalGPT.

NHN Cloud South Korea
AI Researcher Oct 2022 – Nov 2022

- Designed tutorials and evaluation workflows for Korean language models.

KAIST South Korea
Graduate Student Researcher Sep 2020 – Aug 2022

- Researched federated learning, healthcare AI, distribution shifts, and natural language processing under Prof. Edward Choi.
- Served as a teaching assistant for Machine Learning for Healthcare and Programming for AI.

Teaching

KAIST AI612: Machine Learning for Healthcare
Teaching Assistant 2020 – 2021

Led tutorials, managed assignments, and demonstrated core machine-learning implementations.

KAIST AI504: Programming for AI
Teaching Assistant 2020 – 2021

Supported programming tutorials and assignments for classes of up to 100 students.

Community

Co-founder, Virtual Human Lab. Korea-based BioAI research collective connecting clinicians and AI researchers to study human biology with computational methods.

Awards & Scholarships

- Department Prize for Outstanding Student Performance, Yonsei University (2019, 2020; top 3%).
- 3rd Prize, Graduation Capstone Design, Yonsei University (2019).
- 3rd Prize, Industrial Engineering Competition, Korean Ministry of Trade, Industry and Energy (2018).
- Research Assistant Scholarship, KAIST (2020–2022); Full Scholarship, Yonsei University (2017–2019).

Technical Skills

Programming: Python, C/C++ **Machine Learning:** PyTorch, TensorFlow **Tools:** Git, Linux, L^AT_EX